Hugging Face and You:

A hands-on, open-source journey through your ML & NLP development

Steven Kolawole

OSCAFEST; June 16, 2023

LIAR

HANDS-ON

CODE

HUGGING FACE

150 SLIDES

HUGGING FACE

AI

ML

NLP

TRANSFORMERS

HISTORY

ART

LIAR

ANOTHER
LIAR



Wikipedia


LUCID
DELIVERS
FOR COST EFFECTIVE LISP APPLICATIONS
LUCID
LUCID
LUCID
Prime
IBM
apollo
SUN


Rauchen nicht erwünscht
IBM
09



Artificial Intelligence Practitioners

1954 - 1995

1958, Herbert Simon and Allen Newell: ***“Within 10 years a digital computer will be the world’s chess champion.”***







Not in 10 years. Not in 20.

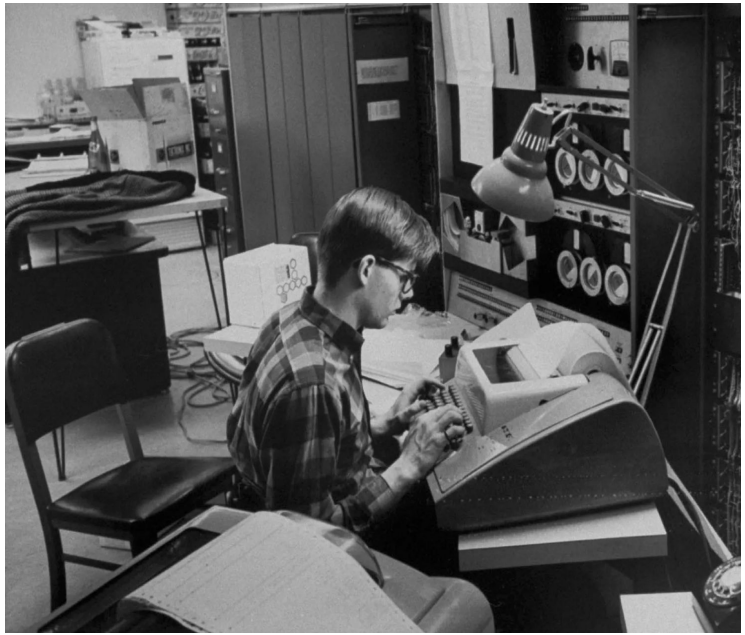
40 years later...



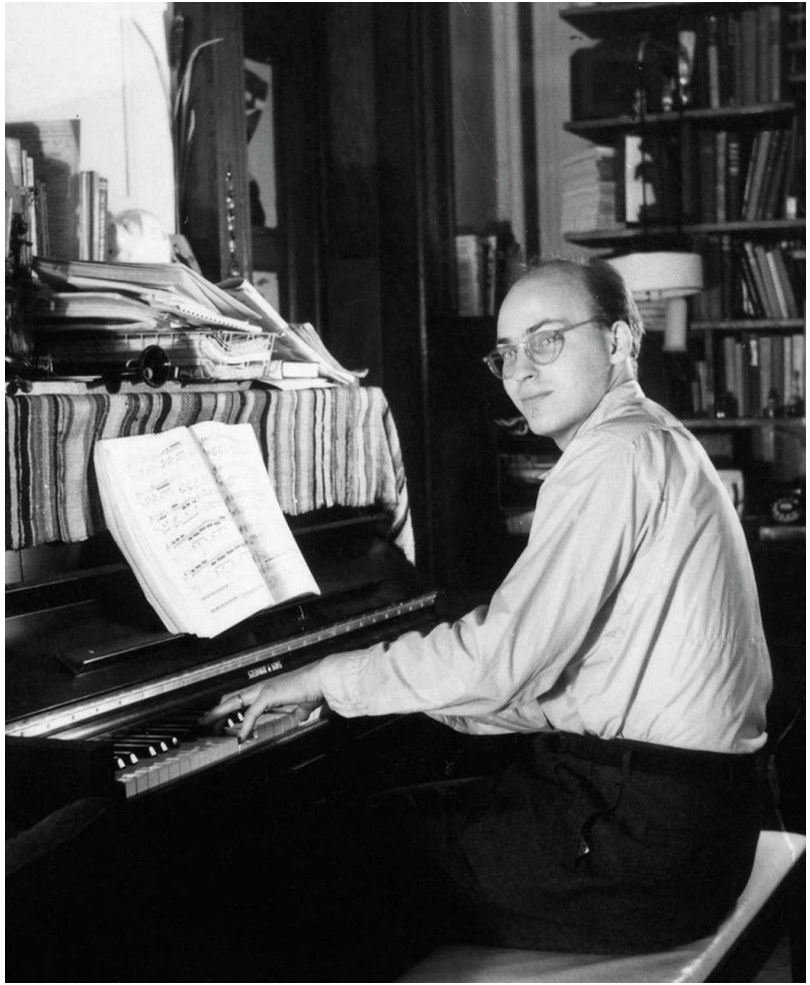
1965, Herbert Simon:
“Machines will be capable, within 20 years, of doing any work a man can do.”

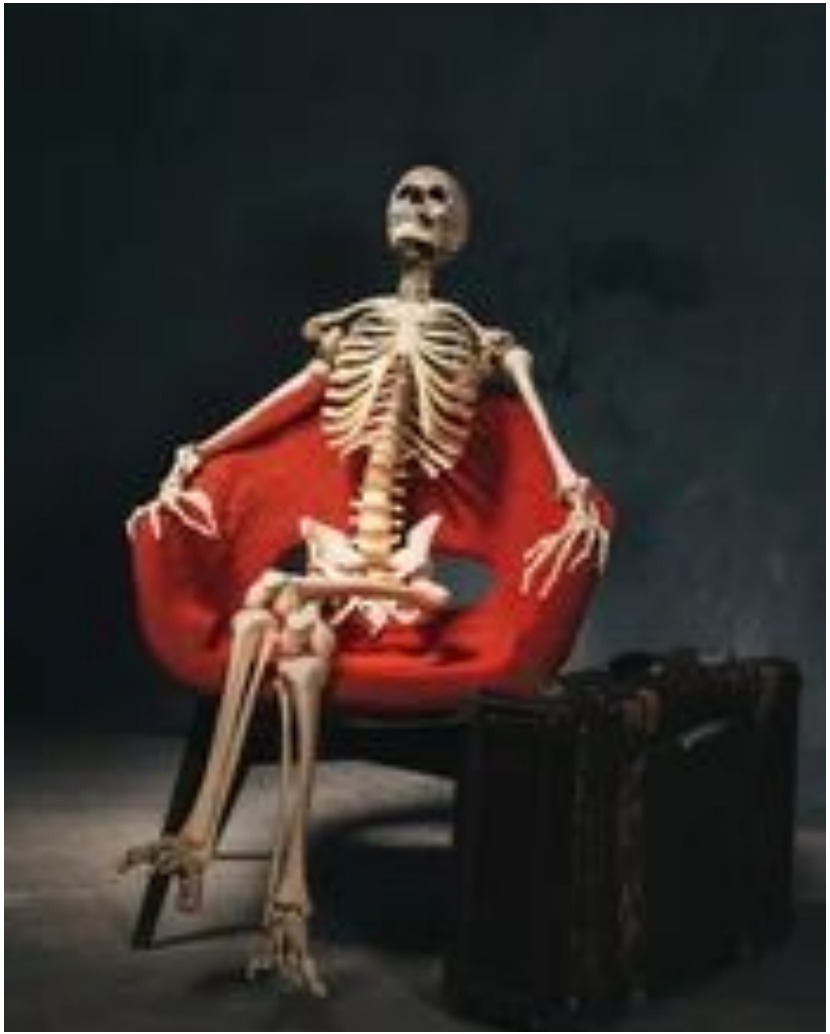


1967, Marvin Minsky: ***“Within a generation, the problem of creating ‘artificial intelligence’ will substantially be solved.”***



1970, Marvin Minsky: ***“In from 3 to 8 years, we will have a machine with the general intelligence of an average human being.”***





NOBEL PRIZE WINNERS
TURING AWARD WINNERS

MILLIONS of USDs

FUNDINGS

EXPERT SYSTEMS
AUTOMATA

Failed



Failed



Failed


HOME
#NIGERIAELECTIONS2023
NEWS
COLUMNS
EDITORIAL
ENTERTAINMENT
POLI
CBN sheds light on cashless policy
by Our Reporter — December 23, 2022 in Latest News, Top News Reading Time: 5 mins read

In 1973 the British AI research community was shocked. A report commissioned by the British Parliament confirmed that the field of research had achieved virtually none of its objectives in recent years. As a result, AI research projects were ceased and funds cut. Not only in Europe, but also in the USA, public funding of civil AI research almost came to a standstill. The US Congress passed a law restricting the funding of basic research without a direct military connection.


came
AI Winter ~~is coming...~~

BUDGETS

CUT

PROJECTS

ABANDONED


MONEY
PLEASE
HOMELESS
PLEASE
HELP
FOR FOOD
BUILD AI



Bottom line: AI is cool to play with in a lab environment, but it will never achieve anything in the real world. *Case closed!*

What went wrong?

What is different now?

DATA

With the dot.com platforms (Google, Facebook) and increased digitalization, we now have petabytes of real-life data

NOT-QUITE, EXPENSIVE SUPERCOMPUTERS

With AWS, GCP, and Azure, we now have cloud computing

CURSE OF DIMENSIONALITY

We have GPUs now for parallel processing

VIBES

We have Afrobeats

VIBES



We have Afrobeats

VIBES



DIFFERENT

CHATGPT

ChatGPT Sprints to One Million Users

Time it took for selected online services to reach one million users



* one million backers ** one million nights booked *** one million downloads

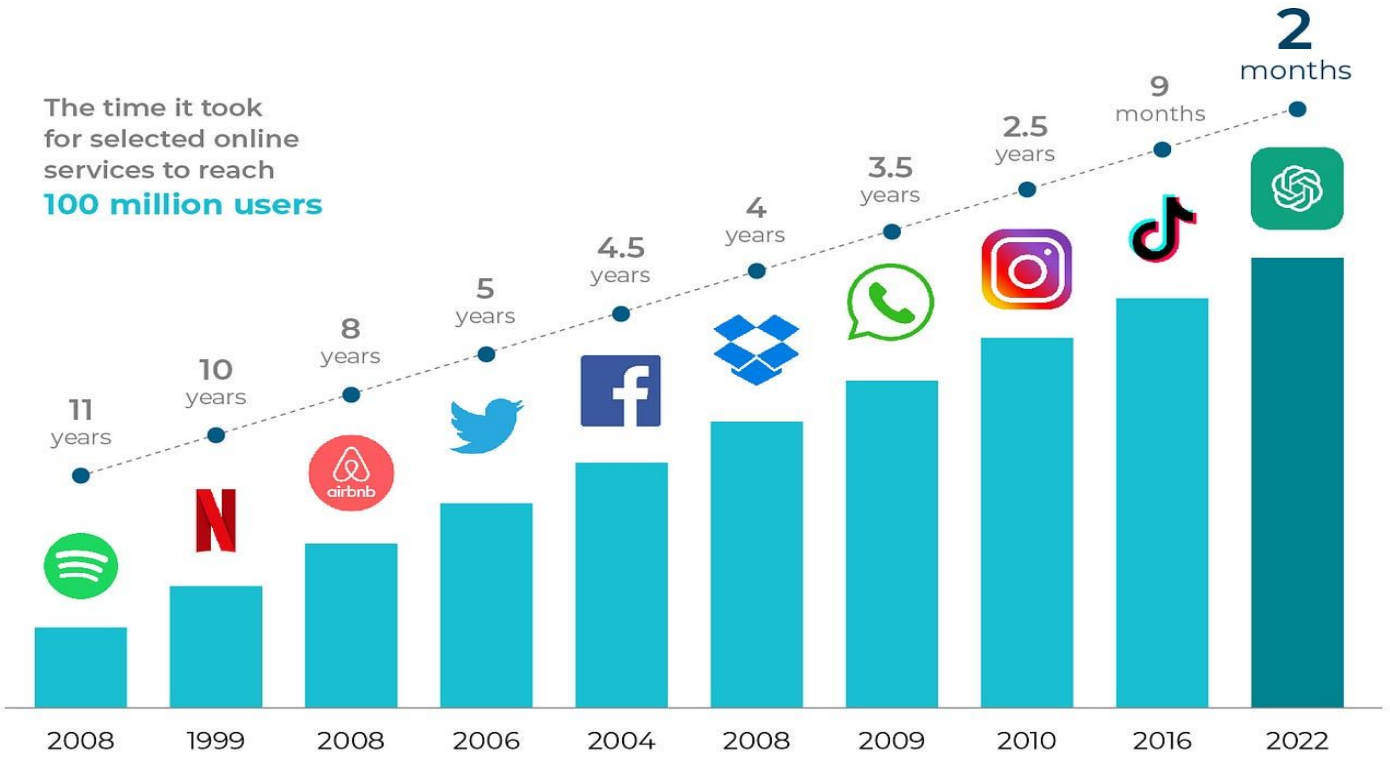
Source: Company announcements via Business Insider/Linkedin

statista

netscribes
RETURN ON KNOWLEDGE

Chat-GPT sprints to 100 million users

The time it took for selected online services to reach **100 million users**



Source: World of Statistics

AI HYPE

FUNDINGS

OPENAI

$27B-$29B

2015

COHERE

$2.2B

2019

ANTHROPIC

$5B-$8B

2021

$113M Seed Funding

Four-week-old **Mistral AI** has just raised **Europe's biggest-ever Seed** round to take on **OpenAI**.

Currently valued at **$260M**, the AI startup plans to launch its first models in 2024.

We're either in a massive **AI bubble** or European startups are getting more **ambitious**.

AI HYPE

Another AI Winter? Or true AI Dominance?

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Independent Researcher, **ML Collective** 2021 - pres.



Steven Kolawole

BSc, **Federal University of Agriculture Abeokuta** 2017 - ~~2021~~ 2023



Steven Kolawole

PhD, **Carnegie Mellon University** 2023 -



Steven Kolawole

Research Engineer, **NewWave**

Hugging Face and You's Workshop

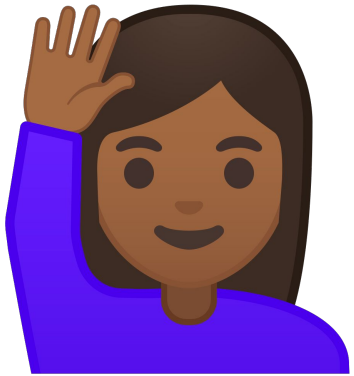
Follow along with the tutorial:

- Slides: https://tinyurl.com/oscafest-huggingface-slides
- Google Colab: https://tinyurl.com/oscafest-huggingface-tutorial
- Twitter: @_stevenkolawole

Questions:

- Ask during the break or after the tutorial

About you



This is my first time hearing / interacting with Artificial Intelligence / Machine Learning.



I know about AI / ML to some extent, but I don’t write code.



I am a programmer but I am not very familiar with AI / ML.

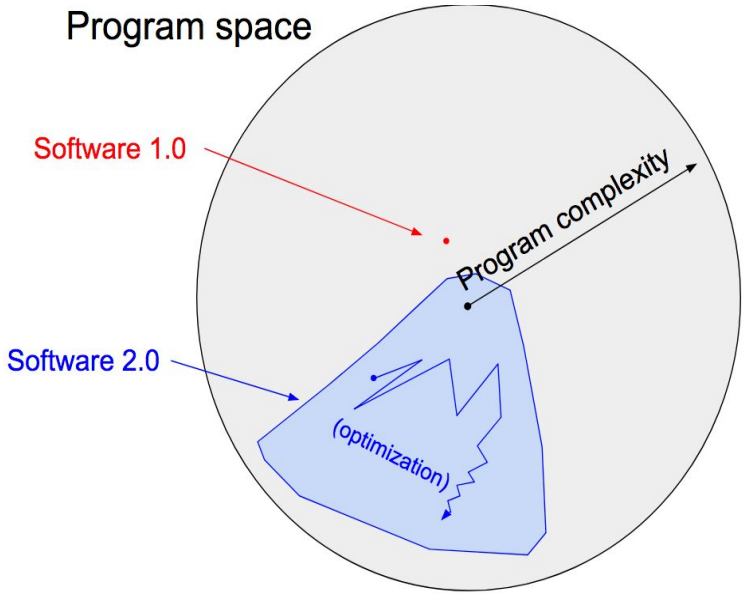


I am comfortable with ML to a good extent, code or maths



Workshop Outline

1. Software 2.0: ML, NLP, & LLMs trends
2. Intro into HuggingFace's Servings
3. Using Spaces
4. A hands-on walkthrough



Traditional Programming vs Machine Learning



Image from LLMBC

Types of Machine Learning

Unsupervised Learning

Learn structure of data to generate more data

"This product does what it is supposed __"

Supervised Learning

Learn how data maps to labels to recognize or predict



→ *cat*



→ *"Hey Siri"*

Reinforcement Learning

Learn how to act in an environment to obtain reward



Image from LLMBC

Inputs and outputs are always just numbers

	Input	Output
What we see		"Lincoln"
What the machine "sees"		[76, 105, 110, 99, 111, 108, 110]

Image from LLMBC

Why is this hard?

- Infinite variety of inputs can all mean the same thing
- Meaningful differences can be tiny
- Structure of the world is complex



"I loved this movie"

"As good as The Godfather"

"🔥 no cap"



Image from LLMBC

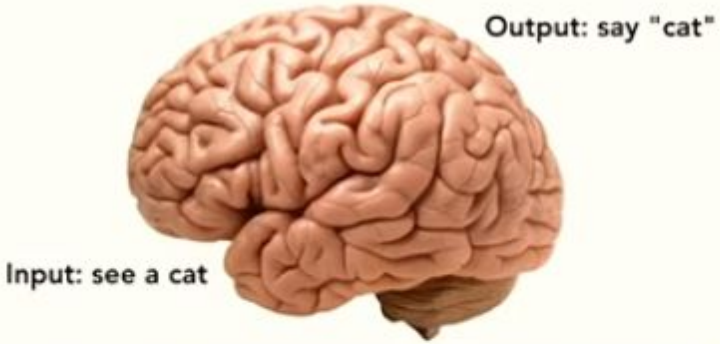
How is it done?

- Linear Regression
- Support Vector Machines (SVM)
- Decision Trees
- Gradient Boosting Algorithms

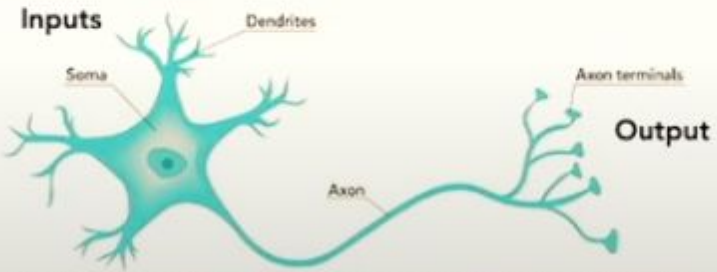
→ But one is dominant in recent days;

- Neural networks (a.k.a. Deep Learning)

Inspiration



https://www.the-scientist.com/the-nutshell/what-made-human-brains-so-big-36663



https://medicalxpress.com/news/2018-07-neuron-axons-spindly-theyre-optimizing.html

- Inspired by what we know to be intelligent: the **brain**
- The brain is composed of billions of **neurons**
- Each neuron receives electrical **inputs** and sends an electrical **output**
- The brain itself has high-level inputs and outputs

Image from LLMBC

Formalization

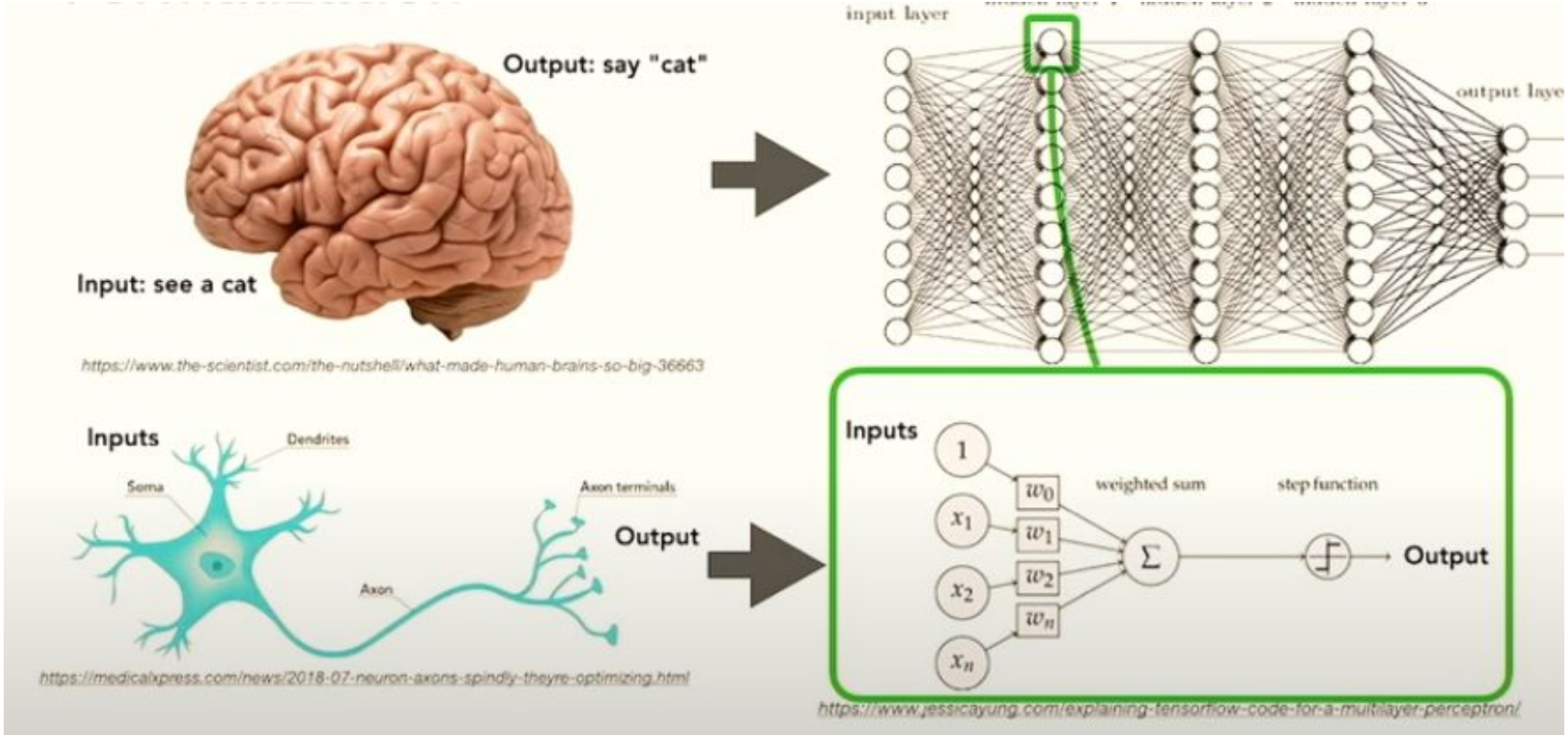
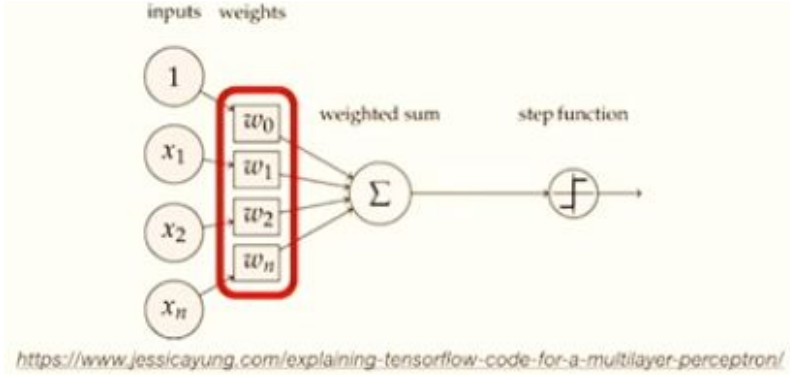
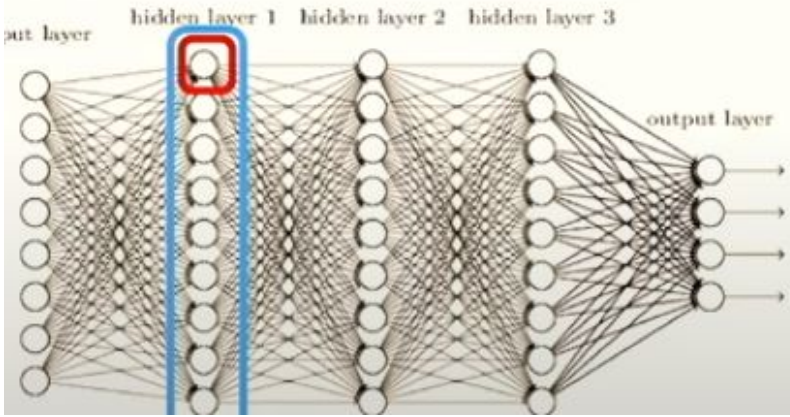


Image from LLMBC

A perceptron is a vector of numbers; a “layer” is a matrix of vector; a neural network is a set of matrices.



https://www.jessicayung.com/explaining-tensorflow-code-for-a-multilayer-perceptron/

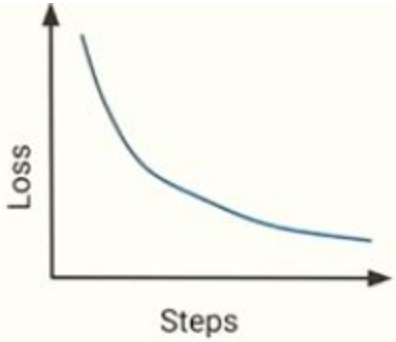



```
[0.58  0.841 0.835 0.18 ]
[0.405 0.813 0.309 0.562]
[0.422 0.229 0.46  0.152]
[0.673 0.429 0.441 0.243]
[0.9   0.744 0.234 0.856]
[0.971 0.486 0.175 0.248]
[0.258 0.588 0.478 0.266]
[0.236 0.496 0.077 0.557]
[0.413 0.322 0.372 0.741]]
```

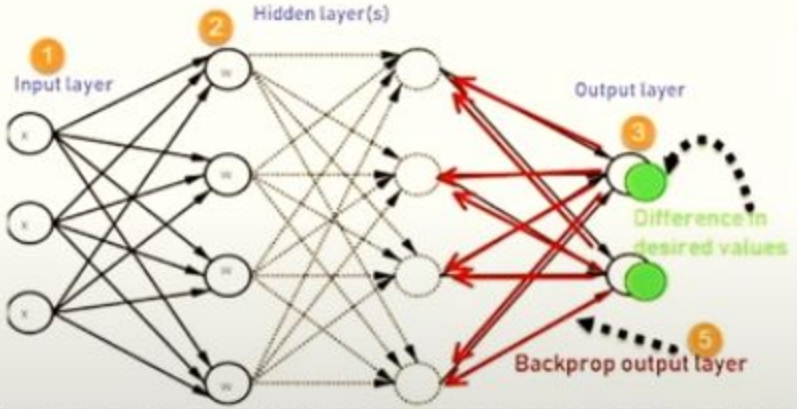
Image from LLMBC

Training

$$loss_{CE} = -\sum(y_i * \log(y'_i))$$



- Data X (e.g. images), labels y (e.g. labels)
- Take a little batch of data x:
  - Use the current model to make a prediction x -> y'
  - Compute loss(y, y')
  - Back-propagate the loss through all the layers of the model
- Repeat until loss stops decreasing



https://www.guru99.com/backpropogation-neural-network.htm

Image from LLMBC

Dataset Splitting

- Split (X, y) into training (~80%), validation (~10%), and test (~10%) sets
- Validation set is for
  - ensuring that training is not "overfitting"
  - setting hyper-parameters of the model (e.g. number of parameters)
- Test set is for measuring validity of predictions on new data

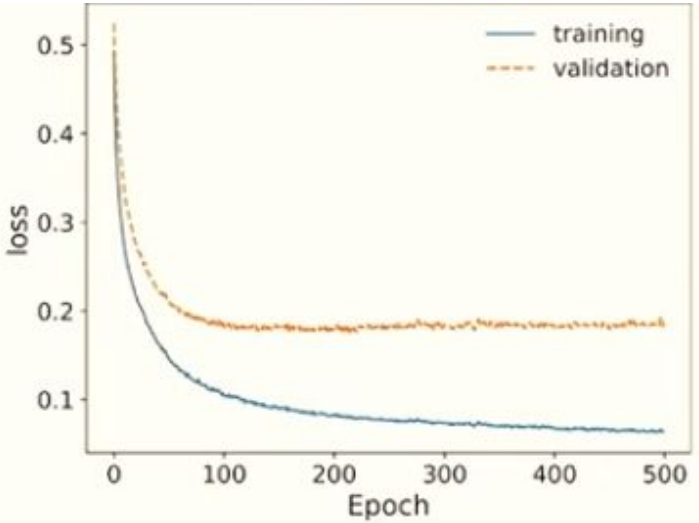


Image from LLMBC



Image from LLMBC

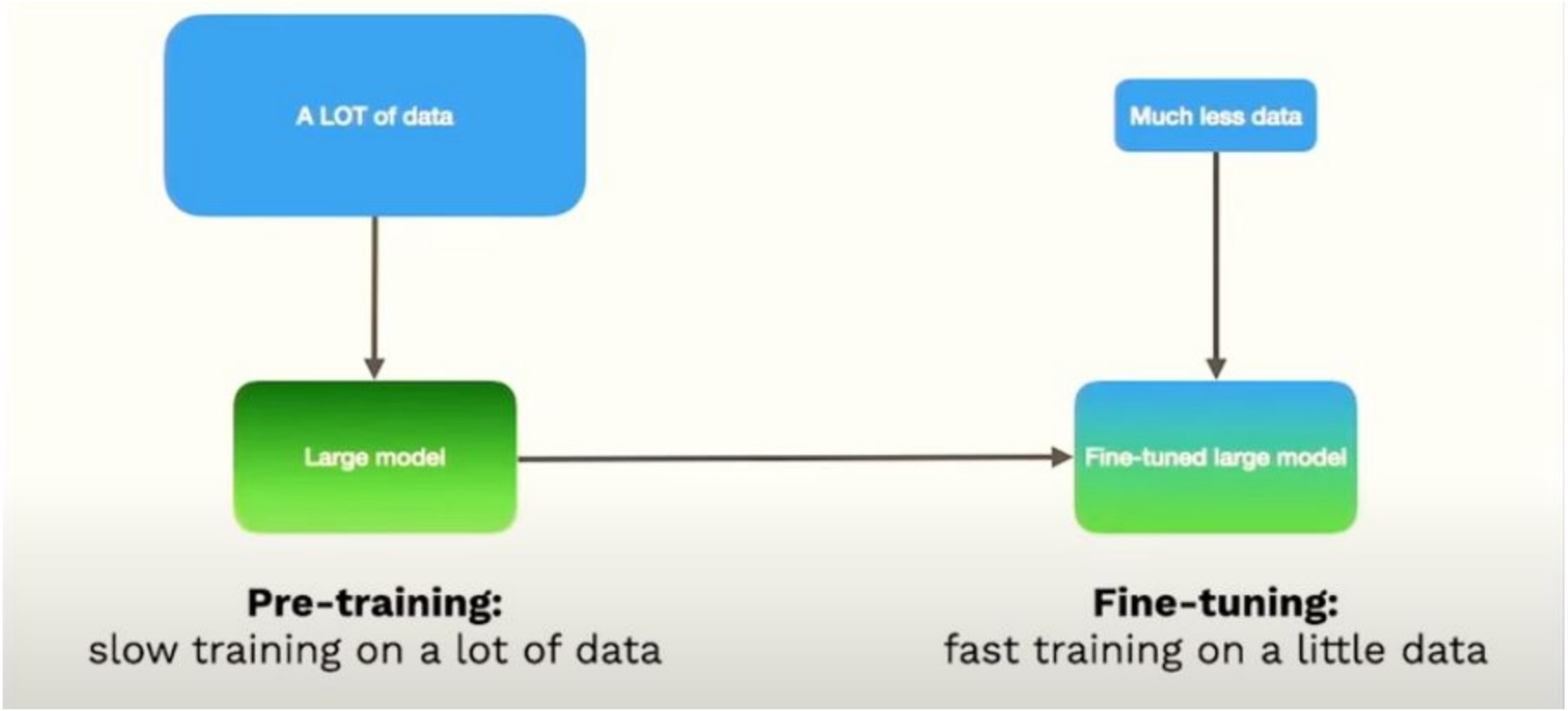


Image from LLMBC

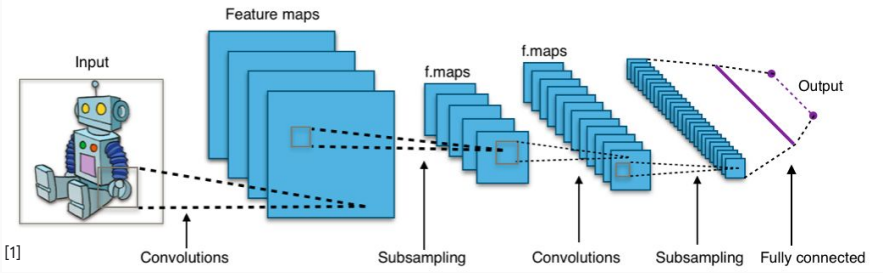
The classic landscape:

One architecture per "community"

http://lucasb.eyer.be/transformer

Computer Vision

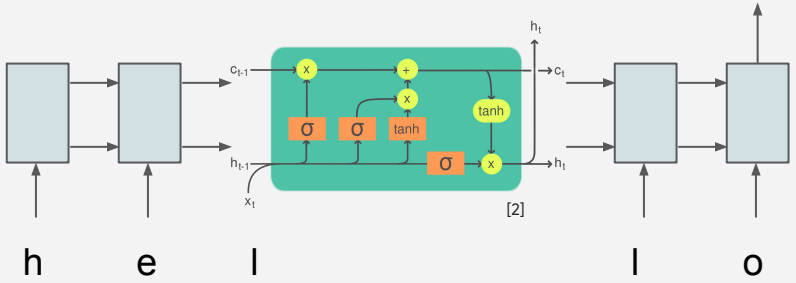
Convolutional NNs (+ResNets)



[1]

Natural Lang. Proc.

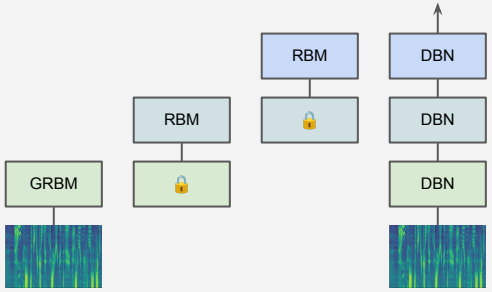
Recurrent NNs (+LSTMs)



h e l l o

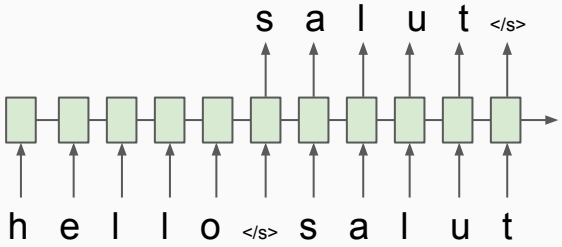
Speech

Deep Belief Nets (+non-DL)



Translation

Seq2Seq



RL

BC/GAIL

Algorithm 1 Generative adversarial imitation learning

Input: Expert trajectories $\tau_E \sim \pi_E$, initial policy and discriminator parameters θ_0, w_0
for $i = 0, 1, 2, \ldots$ **do**
  Sample trajectories $\tau_i \sim \pi_{\theta_i}$
  Update the discriminator parameters from w_i to w_{i+1} with the gradient

$$\hat{\mathbb{E}}_{\tau_i}[\nabla_w \log(D_w(s,a))] + \hat{\mathbb{E}}_{\tau_E}[\nabla_w \log(1 - D_w(s,a))] \quad (17)$$

  Take a policy step from θ_i to θ_{i+1}, using the TRPO rule with cost function $\log(D_{w_{i+1}}(s,a))$. Specifically, take a KL-constrained natural gradient step with

$$\hat{\mathbb{E}}_{\tau_i}[\nabla_\theta \log \pi_\theta(a|s) Q(s,a)] - \lambda \nabla_\theta H(\pi_\theta), \quad \text{where } Q(\bar{s},\bar{a}) = \hat{\mathbb{E}}_{\tau_i}[\log(D_{w_{i+1}}(s,a)) \mid s_0 = \bar{s}, a_0 = \bar{a}] \quad (18)$$

end for

[1] CNN image CC-BY-SA by Aphex34 for Wikipedia https://commons.wikimedia.org/wiki/File:Typical_cnn.png
[2] RNN image CC-BY-SA by GChe for Wikipedia https://commons.wikimedia.org/wiki/File:The_LSTM_Cell.svg

http://lucasb.eyer.be/transformer

The Transformer's takeover:

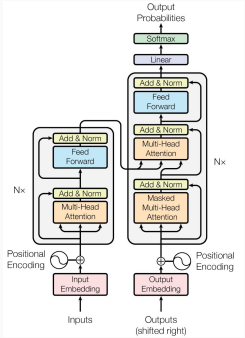
One community at a time

The Transformer's takeover:

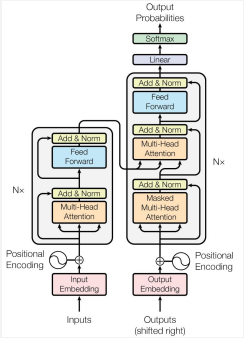
One community at a time

“One technique to rule them all”

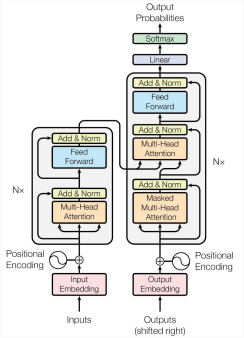
Computer Vision



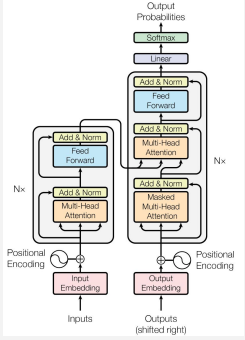
Natural Lang. Proc.



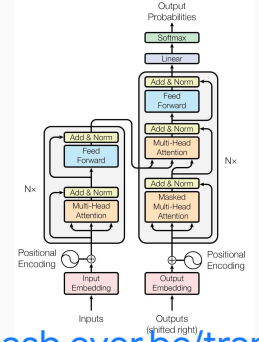
Reinf. Learning



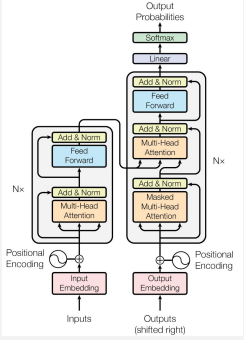
Speech



Translation



Graphs/Science



http://lucasb.eyer.be/transformer

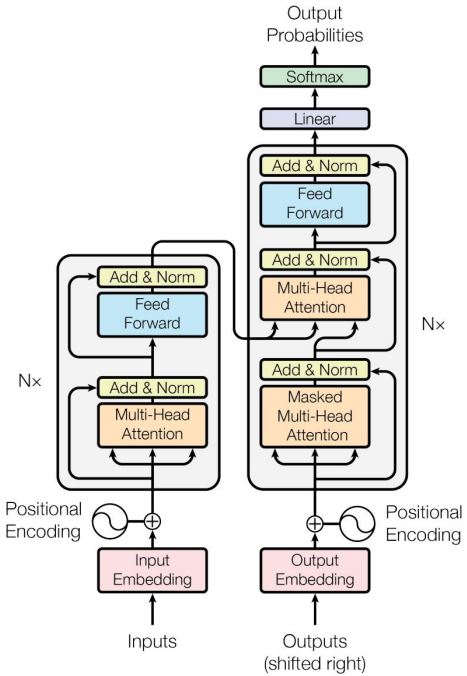
Transformer image source: "Attention Is All You Need" paper





Attention Is All You Need - The Transformer architecture

2017, Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin



Ground-breaking architecture that set SOTA on first translation and later all other NLP tasks.

Thanks to Basil Mustafa for slide inspiration

Transformer image source: "Attention Is All You Need" paper

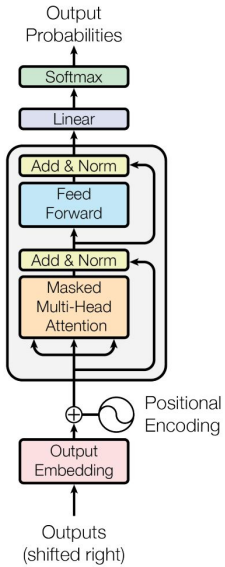
Attention Is All You Need - The Transformer architecture

2017, Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin



Decoder-only GPT

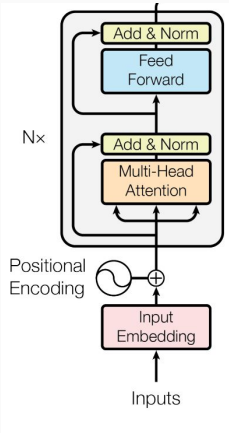
[sat_]



[START] [The_] [cat_]

Encoder-only BERT

[*] [*] **[sat_]** [*] **[the_]** [*]



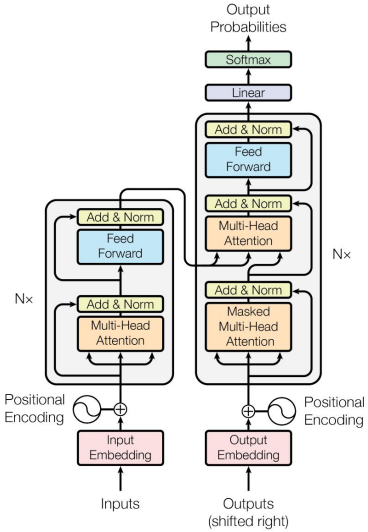
[The_] [cat_] **[MASK]** [on_] **[MASK]** [mat_]

Enc-Dec T5

Das ist gut.

A storm in Attala caused 6 victims.

This is not toxic.



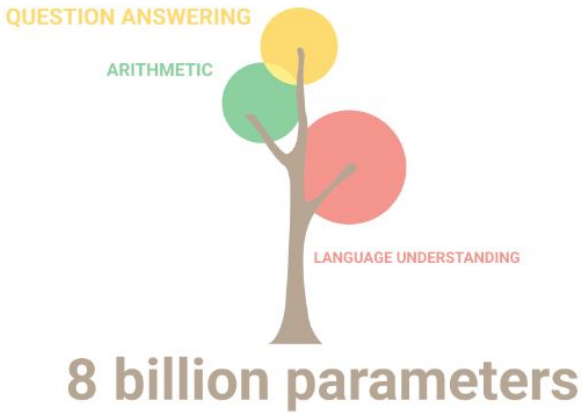
Translate EN-DE: This is good.

Summarize: state authorities dispatched…

Is this toxic: You look beautiful today!

Transformer image source: "Attention Is All You Need" paper

Large Models Develop Increasing NLU Capabilities



Credit: Google AI Blog

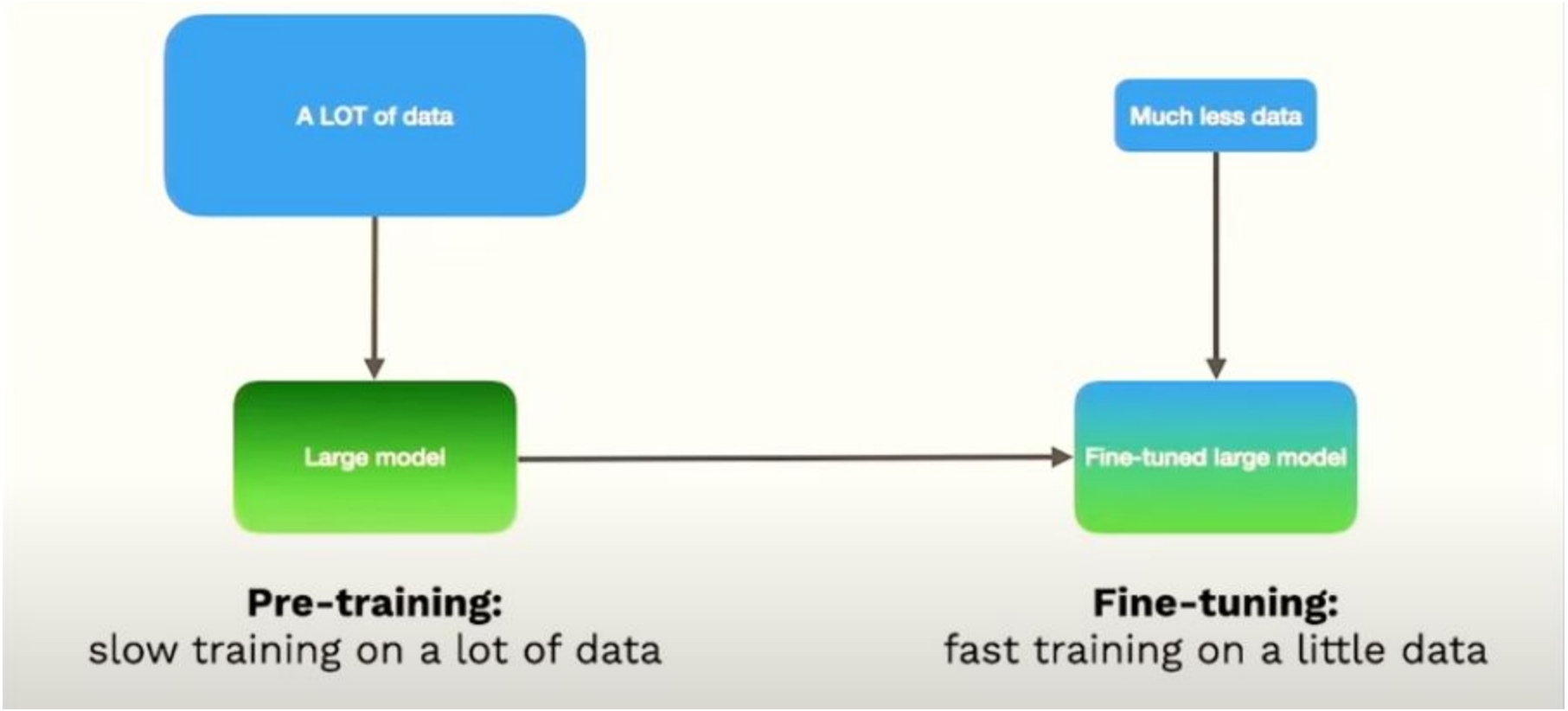
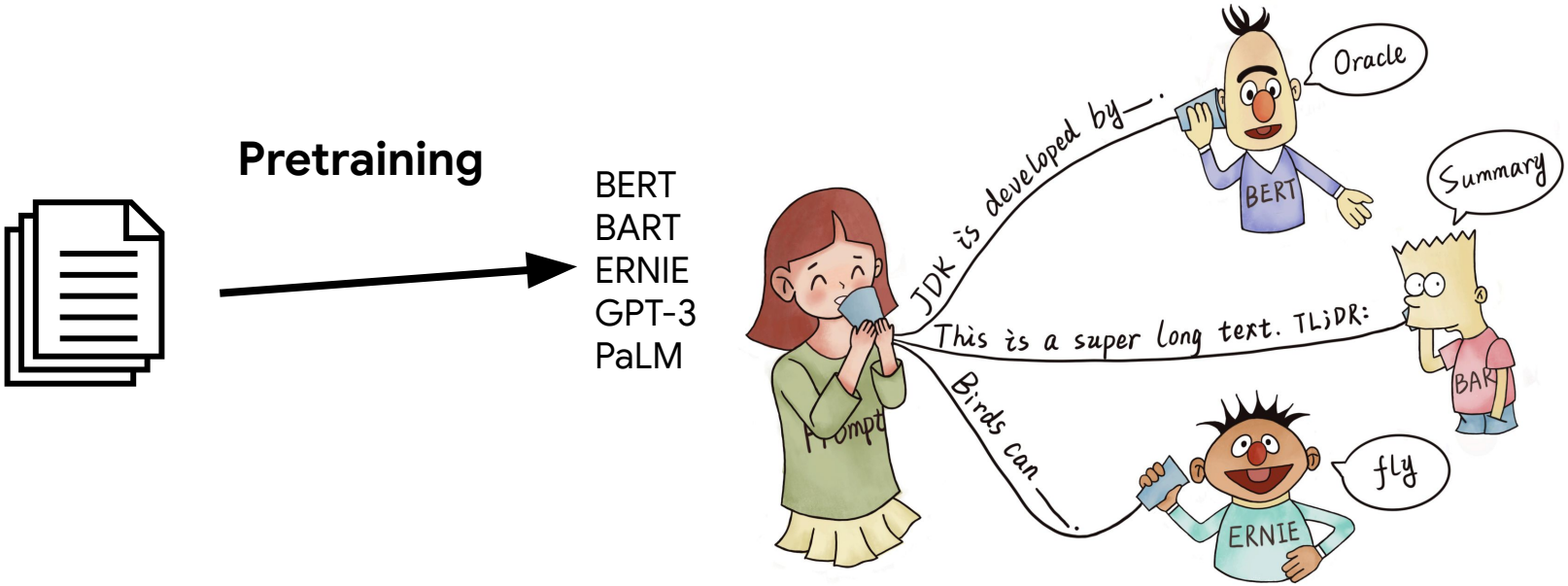


Image from LLMBC

Transfer Learning in the Era of Large Models



Credit: Liu et al. (2021)

Workshop Outline

1. Software 2.0: ML, NLP, & LLMs trends

2. Intro into HuggingFace's Servings

3. Using Spaces

4. A hands-on walkthrough



Hugging Face is a company that provides open-source tools for ML and NLP development. Huggingface's tools are used by a wide variety of developers, including researchers, engineers, and students.

Supported libraries:
https://huggingface.co/docs/hub/models-libraries

Hugging Face's open-source tools include:

1. **The Transformers library**: a library that provides pre-trained models for a variety of NLP tasks.

2. **Datasets**: Huggingface provides a number of datasets that can be used for ML and NLP development.

3. **Tokenizers**: Huggingface provides a number of tokenizers that can be used to preprocess text data for ML and NLP models.

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“Real artists ship.”



python

```python
programmer = "Artist"
```

python

```python
class Programmer:
    ...

    Artist = True
```

“Real artists ship.”

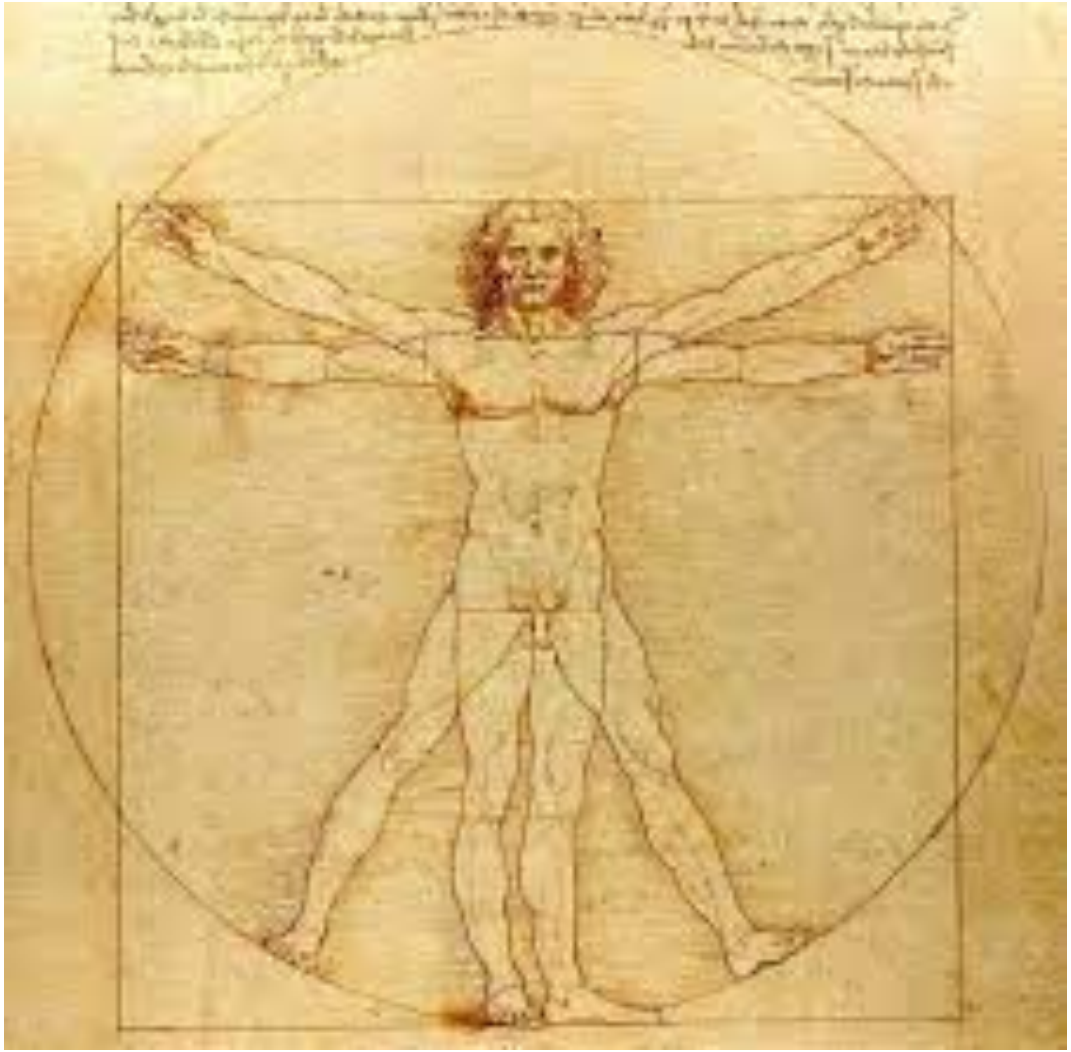


- Artist
- Painter
- Sculptor
- Scientist
- Innovator



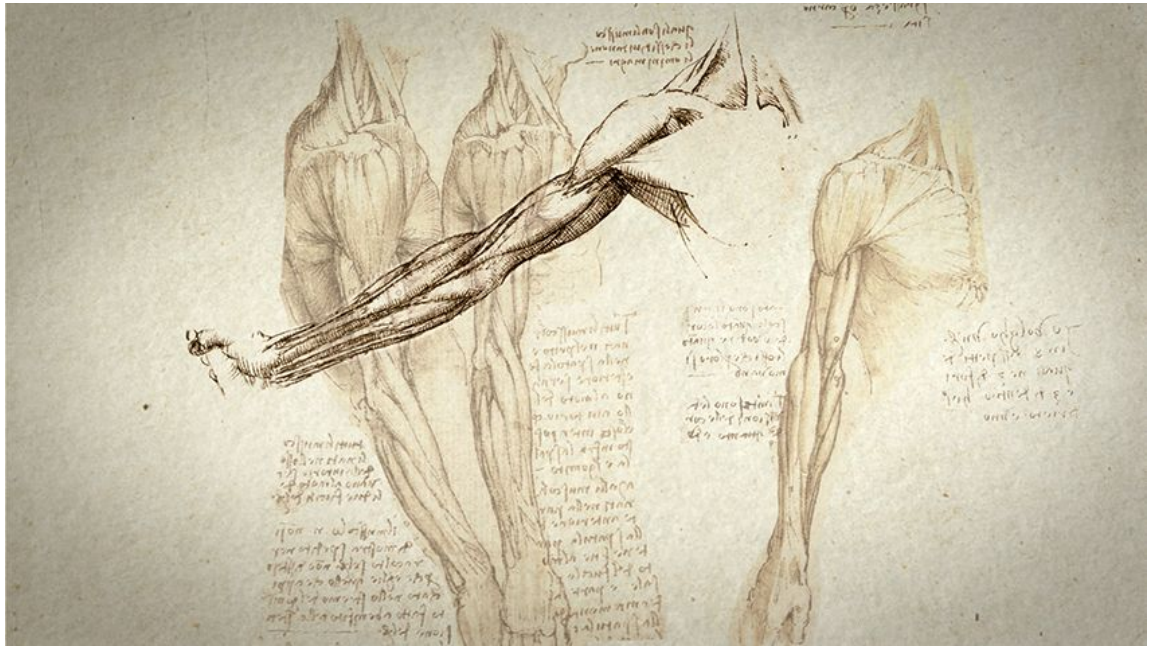




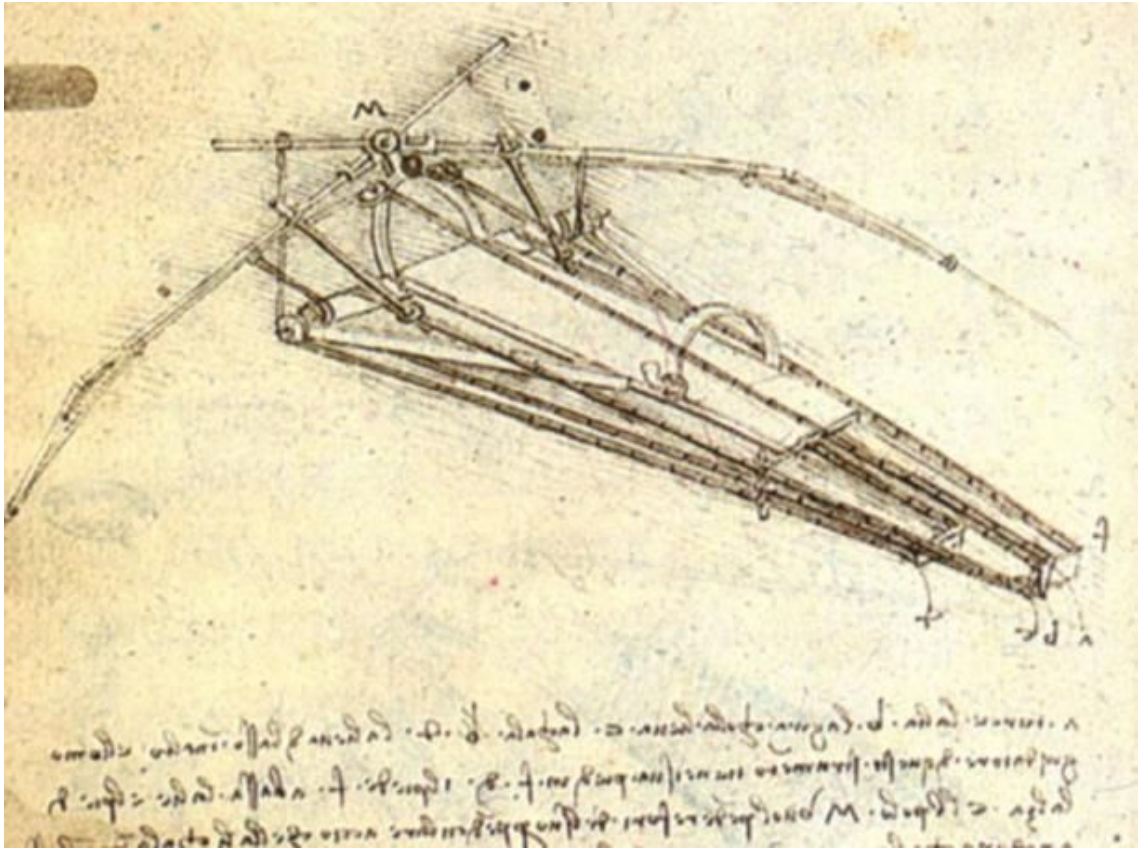


Leonardo da Vinci’s notebooks contain plans for hundreds of technologies common today: machine guns, diving suits, construction cranes, robots, flying machines.

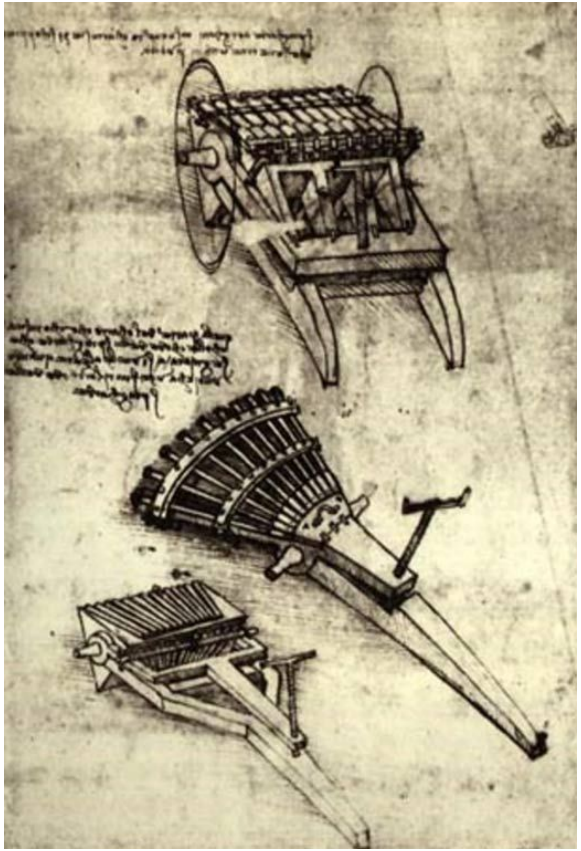
Leonardo da Vinci’s notebooks, written in a secretive reverse script, went unpublished for more than 400 years.



Leonardo da Vinci’s anatomical illustrations, as precise as his technical drawings of machines, were unequaled in accuracy until the photographic techniques of the 19th century. But they were never published in his lifetime.

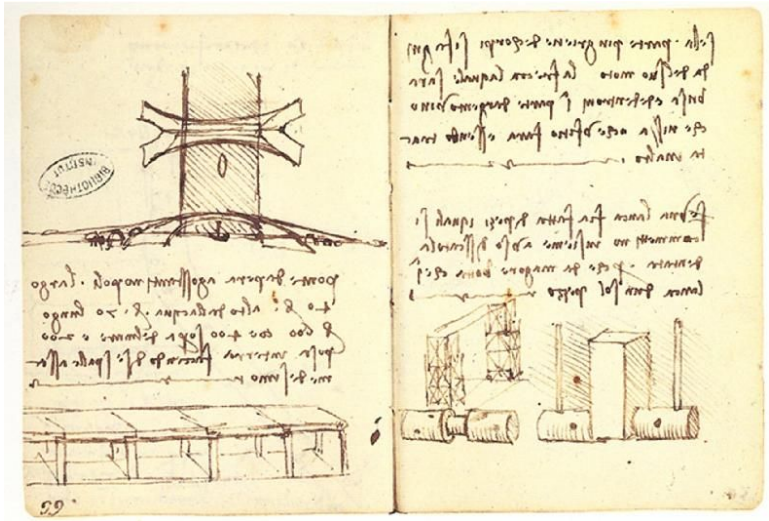


Flying machine - not built until 1903 by the Wright brothers



33 Multi-barrel gun, precursor to machine gun





Designed: 1502

Built: 2001

“Real artists ship.”





Designed: 1502

Built: 2001

Release early, Release often

Failed

Wasted Talent?

Wasted Genius?

!indictment

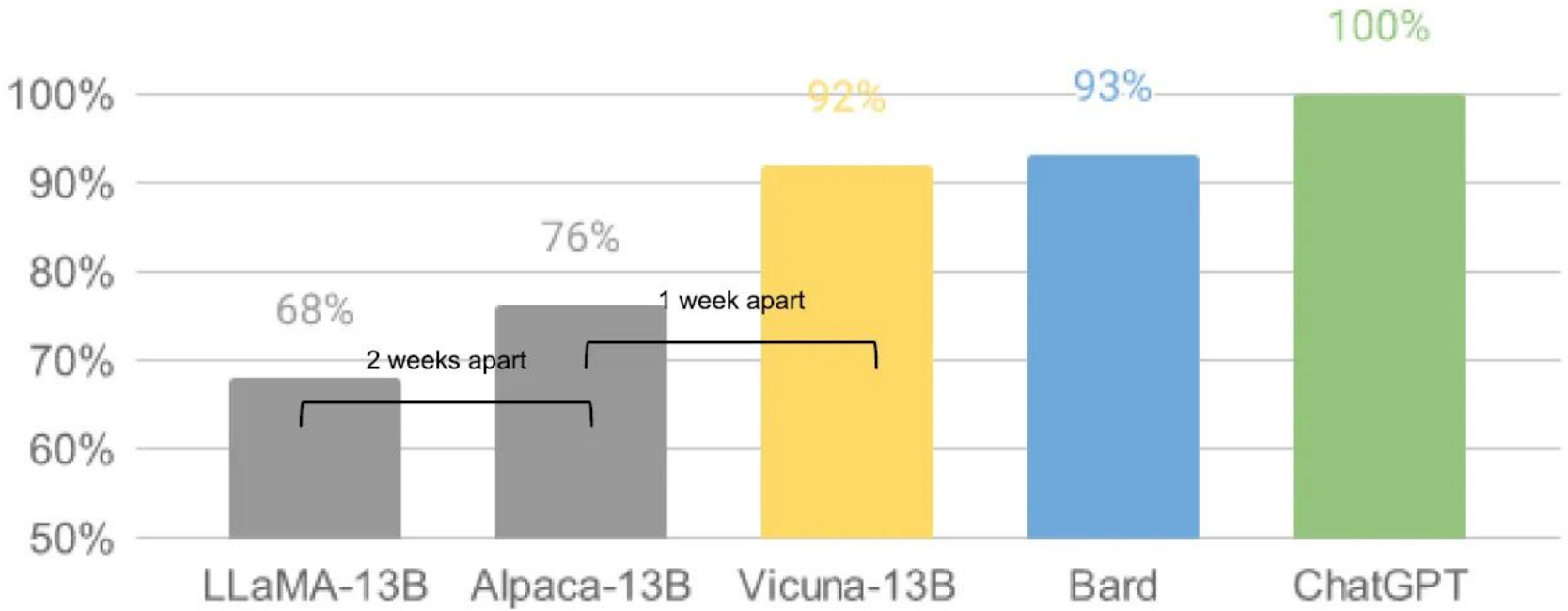
Research

Production

In the current ML landscape,

Research !far Production

Hence, we can innovate, and ship quickly.



*GPT-4 grades LLM outputs. Source: https://vicuna.lmsys.org/

https://www.semianalysis.com/p/google-we-have-no-moat-and-neither

Feb 24, 2023 - LLaMA is Launched

Meta launches LLaMA, open sourcing the code, but not the weights. At this point, LLaMA is not instruction or conversation tuned. Like many current models, it is a relatively small model (available at 7B, 13B, 33B, and 65B parameters) that has been trained for a relatively large amount of time, and is therefore quite capable relative to its size.

March 3, 2023 - The Inevitable Happens

Within a week, LLaMA is leaked to the public. The impact on the community cannot be overstated. Existing licenses prevent it from being used for commercial purposes, but suddenly anyone is able to experiment. From this point forward, innovations come hard and fast.

https://www.semianalysis.com/p/google-we-have-no-moat-and-neither

March 12, 2023 - Language models on a Toaster

A little over a week later, Artem Andreenko gets the model working on a Raspberry Pi. At this point the model runs too slowly to be practical because the weights must be paged in and out of memory. Nonetheless, this sets the stage for an onslaught of minification efforts.

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March 13, 2023 - Fine Tuning on a Laptop

The next day, Stanford releases Alpaca, which adds instruction tuning to LLaMA. More important than the actual weights, however, was Eric Wang's alpaca-lora repo, which used low rank fine-tuning to do this training "within hours on a single RTX 4090".

Suddenly, anyone could fine-tune the model to do anything, kicking off a race to the bottom on low-budget fine-tuning projects. Papers proudly describe their total spend of a few hundred dollars. What's more, the low rank updates can be distributed easily and separately from the original weights, making them independent of the original license from Meta. Anyone can share and apply them.

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March 18, 2023 - Now It's Fast

Georgi Gerganov uses 4 bit quantization to run LLaMA on a MacBook CPU. It is the first "no GPU" solution that is fast enough to be practical.

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March 19, 2023 - A 13B model achieves “parity” with Bard

The next day, a cross-university collaboration releases Vicuna, and uses GPT-4-powered eval to provide qualitative comparisons of model outputs. While the evaluation method is suspect, the model is materially better than earlier variants. **Training Cost: $300.**

Notably, they were able to use data from ChatGPT while circumventing restrictions on its API - They simply sampled examples of “impressive” ChatGPT dialogue posted on sites like ShareGPT.

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March 25, 2023 - Choose Your Own Model

Nomic creates GPT4All, which is both a model and, more importantly, an ecosystem. For the first time, we see models (including Vicuna) being gathered together in one place. **Training Cost: $100.**

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March 28, 2023 - Open Source GPT-3

Cerebras (not to be confused with our own Cerebra) trains the GPT-3 architecture using the optimal compute schedule implied by Chinchilla, and the optimal scaling implied by μ-parameterization. This outperforms existing GPT-3 clones by a wide margin, and represents the first confirmed use of μ-parameterization "in the wild". These models are trained from scratch, meaning the community is no longer dependent on LLaMA.

https://www.semianalysis.com/p/google-we-have-no-moat-and-neither

March 28, 2023 - Multimodal Training in One Hour

Using a novel Parameter Efficient Fine Tuning (PEFT) technique, LLaMA-Adapter introduces instruction tuning and multimodality in one hour of training. Impressively, they do so with just 1.2M learnable parameters. The model achieves a new SOTA on multimodal ScienceQA.

https://www.semianalysis.com/p/google-we-have-no-moat-and-neither

April 3, 2023 - Real Humans Can't Tell the Difference Between a 13B Open Model and ChatGPT

Berkeley launches Koala, a dialogue model trained entirely using freely available data.

They take the crucial step of measuring real human preferences between their model and ChatGPT. While ChatGPT still holds a slight edge, more than 50% of the time users either prefer Koala or have no preference. **Training Cost: $100.**

https://www.semianalysis.com/p/google-we-have-no-moat-and-neither

April 15, 2023 - Open Source RLHF at ChatGPT Levels

Open Assistant launches a model and, more importantly, a dataset for Alignment via RLHF. Their model is close (48.3% vs. 51.7%) to ChatGPT in terms of human preference. In addition to LLaMA, they show that this dataset can be applied to Pythia-12B, giving people the option to use a fully open stack to run the model. Moreover, because the dataset is publicly available, it takes RLHF from unachievable to cheap and easy for small experimenters.

https://www.semianalysis.com/p/google-we-have-no-moat-and-neither

OPEN SOURCE

POWER

GOOD


Awesome

Auto-GPT has more GitHub stars than PyTorch (82K at the moment) and is the “fastest growing GitHub repo in history, eclipsing decade old open source projects in 2 weeks.”

INSANE

GREAT

INSANELY
GREAT

Spaces

Discover amazing ML apps made by the community!

https://huggingface.co/spaces

Task & Break

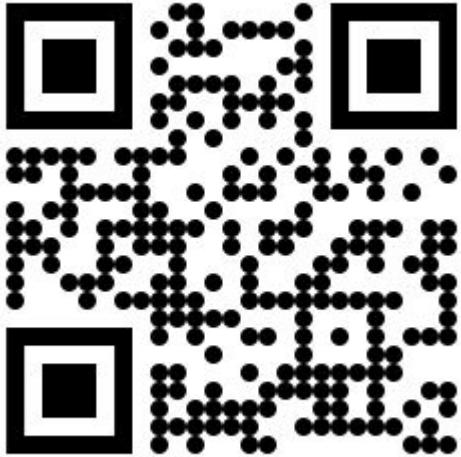
Find the most amazing ML app you can find on Spaces.

https://huggingface.co/spaces

Task & Break

Find the most amazing ML app you can find on Spaces.

https://huggingface.co/spaces



3 volunteers to tell us about the best app they find, and what makes the app super interesting.

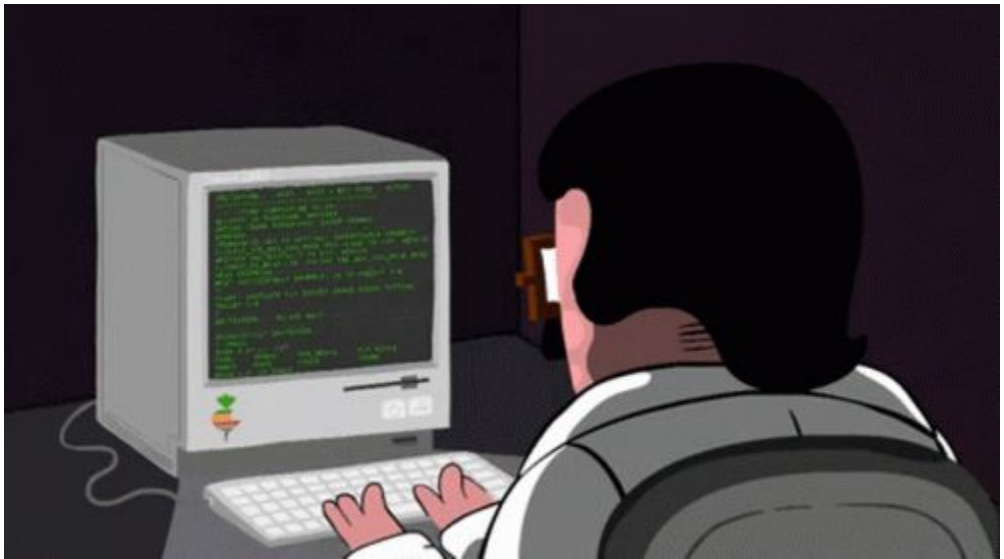
Workshop Outline

1. Software 2.0: ML, NLP, & LLMs trends
2. Intro into HuggingFace's Servings
3. Using Spaces
4. A hands-on walkthrough



Code lab

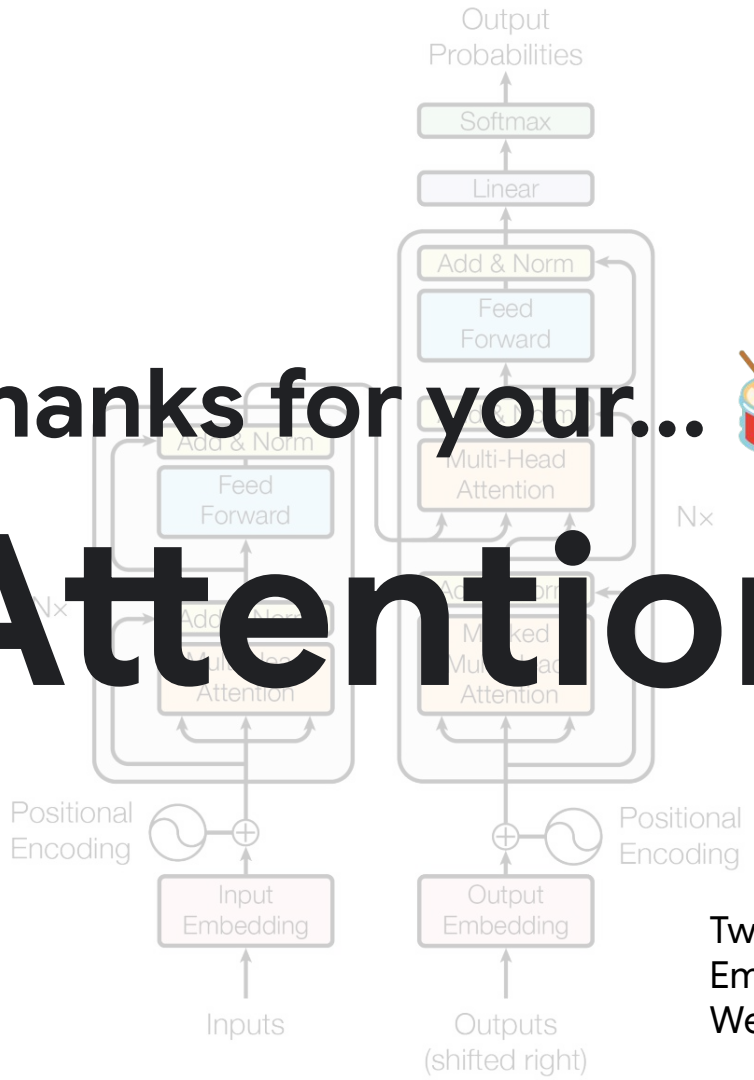
- Google Colab: https://tinyurl.com/oscafest-huggingface-tutorial



Code lab

- Google Colab: https://tinyurl.com/oscafest-huggingface-tutorial



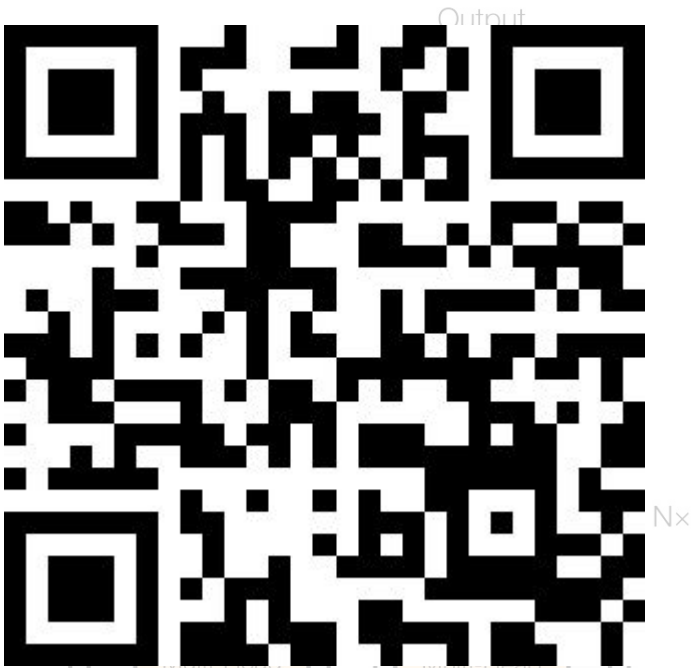


Thanks for your... 🥁

Attention

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Website: stevenkolawole.github.io

Transformer image source: "Attention Is All You Need" paper



https://tinyurl.com/feedback-for-steven

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Email: kolawolesteven99@gmail.com
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